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Exceptional high gas permeability and physical aging resistance of silyl-functionalized fluorene-based poly(substituted acetylene)s

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ABSTRACT

Poly(substituted acetylene)s have emerged as highly promising materials for gas-permeable membranes in gas separation systems. A novel series of silyl-functionalized fluorene-based poly(substituted acetylene)s (Copoly (Flu1-Flu2)s) were synthesized via copolymerization of 1-phenyl-2-(7-trimethylsilyl-9,9-bistrimethylsilylfluoren-2-yl)acetylene (Flu1) with 1-phenyl-2-(7-trimethylsilyl-9,9-dimethylfluoren-2-yl)acetylene (Flu2). Solution-cast copolymer membranes exhibited exceptional gas permeabilities, with the Copoly(Flu1-Flu2)(4:1) membrane showing oxygen (P_{O2} = 19,800 Barrer) permeability that surpass those of homopolymers. Desilylation via trifluoroacetic acid treatment altered permeability depending on the feed ratio, attributed to competitive effects of chain packing densification and microvoid generation. XRD and N_2 adsorption analyses revealed that higher Flu1 content increased chain spacing (d-spacing = 15.2 Å) and BET surface area (492 m^2/g), with desilylation inducing pore structure reorganization. The membranes exhibited good thermal stability (T_d > 420 °C) and resistance to physical aging (P_{O2} reduction <10 % within 240 days), underscoring their potential for gas separation applications.

1. Introduction

Membrane-based gas separation has emerged as an energy-efficient and environmentally benign alternative to conventional technologies, such as cryogenic distillation and adsorption, for critical industrial processes including carbon capture, hydrogen purification, and oxygen/nitrogen enrichment [1–5]. Gas separation membranes necessitate a balance of high permeability and selectivity, although the desired performance varies by application. For instance, oxygen-enriched air production prioritizes permeability, whereas hydrogen purification demands high selectivity. However, few materials meet these requirements, limiting the practical use of gas separation membranes. While polymeric membranes currently dominate commercial applications due to their processability and scalability, their performance is inherently constrained by the permeability-selectivity trade-off, as characterized by the Robeson upper bounds [6]. This fundamental limitation underscores the challenge in developing advanced membranes for industrial applications.

High-free-volume glassy polymers (HFVPs), particularly polymers of intrinsic microporosity (PIMs) and poly(substituted acetylene)s, have garnered significant interest for their exceptional gas permeability arising from inefficient chain packing and microporosity [7–9]. In order to surpass the permeability-selectivity trade-off, the increase in rigidity of polymer chains is one of the important factors. Among HFVPs, poly (substituted acetylene)s exhibit remarkable gas transport properties due to their rigid backbones and tunable side groups that inhibit efficient chain packing [9–11]. Therefore, poly(substituted acetylene)s may have the potential to overcome the trade-off relationship. In addition, poly (substituted acetylene)s exhibit relatively simple repeating structures in comparison to PIMs, which facilitates the regulation of their performance through the steric effects of the side chains.

Fluorene-based polyacetylenes, in particular, stand out because the

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substituent at the 9-position of fluorene extends perpendicularly to the aromatic plane, effectively suppressing π-π stacking and creating substantial free volume [12–14]. Recent studies indicate that silyl functionalization (e.g., trimethylsilyl, $SiMe_3$) further enhances gas permeability by introducing steric bulkiness and high local mobility, facilitating gas diffusion [12–15]. Additionally, silyl groups can be selectively removed via protic acid, a post-polymerization modification that generates microvoids and tailors free volume distribution [9–15]. Moreover, copolymerization of substituted acetylene monomers has also proven effective in achieving high gas permeability, yielding copolymers with superior gas permeability compared to homopolymers [15–17]. This approach offers a straightforward and efficient strategy for enhancing gas permeability.

Despite these advances, critical challenges persist: (1) rapid physical aging in HFVPs leads to declining permeability over time due to chain relaxation [18,19]; (2) achieving synergistically high permeability and aging resistance remains elusive [20,21]; and (3) the impact of copolymer composition and desilylation on chain packing, pore structure, and long-term stability is poorly understood [15–17]. Recent efforts to mitigate aging include incorporating bulky, rigid moieties [22–24] and copolymerization strategies that optimize fractional free volume (FFV) while restricting chain mobility [9]. For fluorene-based polyacetylenes, we reported that polymers bearing silyl groups achieved high O_2 permeability (P_{O2} >1500 Barrer) and suffered >30 % aging loss over 90 days [12]. Thus, designing membranes that concurrently achieve record-high permeability, selectivity, and aging resistance requires precise molecular engineering of backbone rigidity, side-group steric, and post-synthetic pore manipulation.

In this study, we address these challenges through the synthesis and characterization of a novel series of silyl-functionalized fluorene-based acetylene copolymers (**Copoly(Flu1-Flu2)**s and **DScopoly(Flu1-Flu2)**s; Scheme 1), where **Flu1** and **Flu2** monomers feature distinct 9,9-bis(trimethylsilyl)fluorenyl and 9,9-dimethylfluorenyl groups, respectively. By systematically varying the **Flu1**:**Flu2** feed ratio, we leverage the distinct steric effects of their substituents. The bulky 9,9-bis (trimethylsilyl) groups in **Flu1** enhance free volume and permeability, whereas the 9,9-dimethyl groups in **Flu2** introduce steric hindrance to restrict excessive chain mobility. This balance prevents the rapid relaxation observed in **Flu1**-rich homopolymers and avoids the low permeability of **Flu2**-rich homopolymers. Furthermore, controlled desilylation selectively removes trimethylsilyl groups, reorganizing the pore structure. In **Flu2**-rich copolymers, this process generates additional microvoids without compromising chain rigidity, as the methyl groups maintain sufficient steric hindrance to resist packing. This dual strategy enables the membranes to achieve ultrahigh permeability while exhibiting minimal permeability loss, thereby establishing a new paradigm for designing high-performance membranes that simultaneously address the permeability-aging resistance trade-off, advancing the frontier of polymeric materials for energy-efficient gas separations.

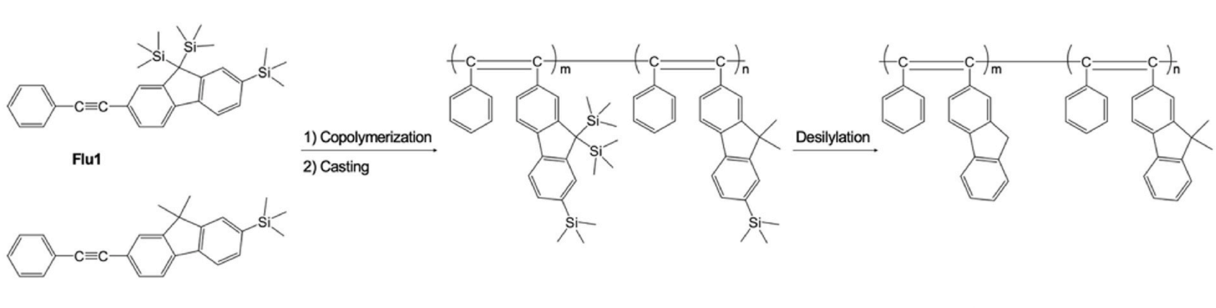
2. Experimental

2.1. Materials

Bromine (>98 %) was purchased from Tokyo Chemical Industry Co., Ltd. Fluorene (98 %), chlorotrimethylsilane (99 %), *n*-butyllithium (2.5 M solution in hexane), diisopropylamine (≥99 %), phenylacetylene (>97 %), methyl iodide (>98 %), potassium iodide (99 %), potassium hydroxide (99.999 %), copper(I) iodide (99 %), dichlorobis(triphenylphosphine)palladium(II) (98 %), triphenylphosphine (99 %), trifluoroacetic acid (TFA, 99 %), tantalum(V) chloride (99.999 %), and other organic solvents were purchased from Shanghai Adamas Reagent Co., Ltd. Sodium thiosulfate (≥97 %) and calcium hydride (≥97 %) were purchased from Shanghai Titan Technology Co., Ltd. The above chemicals were used as received without further purification. *n*-Tetrabutyltin (95 %, Adamas) and cyclohexane (99.9 %, Adamas) were used after distillation over calcium hydride. 1-Phenyl-2-(7-trimethylsilyl-9,9-bistrimethylsilyl-fluoren-2-yl)acetylene (**Flu1**) [12] and 1-phenyl-2-(7-trimethylsilyl-9,9-dimethyl-fluoren-2-yl)acetylene (**Flu2**) [13] were synthesized according to the previously reported methods.

2.2. Characterization and methods

The Fourier transform infrared (FT-IR) spectra of the membranes were obtained using attenuated total reflection (ATR) mode using a Nicolet iS20 spectrophotometer (Thermo Fisher Scientific). The nuclear magnetic resonance (NMR) spectra were acquired using a JEOL ECX-500 spectrometer (500 MHz) at room temperature, with $CDCl_3$ as the solvent. Thermal gravimetric analysis (TGA) was used to analyze the thermal stability of membranes by the TG-DTA 8078G1 of Rigaku instrument at a heating rate of 10 °C/min from 30 °C to 800 °C under a nitrogen atmosphere. Gel permeation chromatography (GPC) analysis of the copolymers was performed on the Waters 1515 calibrated against polystyrene standards to obtain the molecular weight, using tetrahydrofuran (THF) as eluent at a flow rate of 1 mL/min, and the polymer dispersion index (PDI) was calculated by PDI = M_w/M_n. The X-ray diffraction (XRD) was used to investigate the chain structure and average chain spacing of the polymer membranes. The Bruker D8 ADVANCE instrument was utilized with Cu radiation of wavelength (λ) 1.54 Å. The value of the d-spacing was calculated by means of Bragg's law ($d = \lambda/2\sin\theta$), using θ of the broad peak maximum. The BET surface areas were obtained through N_2 adsorption testing using the Nova 800 (Anton Paar Instruments) at 77 K. Pore size distributions were calculated using the Barrett-Joyner-Halenda (BJH) method. Samples of membranes were degassed at 120 °C for 12 h before testing. The surface morphology and elemental composition of the membranes were characterized using a Hitachi SU8600 Scanning Electron Microscope (SEM) equipped with an energy-dispersive X-ray spectroscopy (EDS) detector. Samples were sputter-coated with a thin layer of platinum (≈5 nm) prior to analysis to minimize charging effects and enhance image quality. The gas permeability (P) of membranes was measured with a differential pressure gas permeameter (GTR-G1, Pubtester Instrument, Jinan, China) at 25 °C



Scheme 1. Preparation of fluorene-based poly(disubstituted acetylene)s.

under a 1 atm upstream pressure and a 3 Pa downstream pressure with the gases of CO_2, O_2, and N_2, successively. Prior to gas permeability measurements, membranes were vacuum-dried at 60 °C for 12 h to remove adsorbed moisture and residual solvents. *P* was calculated from the slope in the steady-state region of the time-pressure curve and is quoted in Barrer (1 Barrer = 10^{-10} cm^3 (STP) cm cm^{-2} s^{-1} $cmHg^{-1}$). Effective area of membranes for permeability tests was 7 cm^2, with 3 replicate measurements per membrane (ca. RSD <0.8 %).

2.3. Syntheses of the **Copoly(Flu1-Flu2)** copolymers

The **Copoly(Flu1-Flu2)** copolymers were synthesized via copolymerizations with varying feed ratios of the **Flu1** and **Flu2** monomers. The copolymerizations were carried out under the following conditions: $[\mathbf{Flu1} + \mathbf{Flu2}]_0 = 0.20$ M, $[TaCl_5] = 0.020$ M, and $[n\text{-}Bu_4Sn] = 0.040$ M. For example, to prepare **Copoly(Flu1-Flu2)(1:4)**, a monomer solution consisting of **Flu1** (0.48 g), **Flu2** (0.37 g), and cyclohexane (5.0 mL) was prepared in a Schlenk tube under dry nitrogen. In another tube, a catalytic solution of cyclohexane (4.0 mL), $TaCl_5$ (0.14 g), and n-Bu_4Sn (0.8 M cyclohexane solution, 1.0 mL) was also prepared under dry nitrogen and stirred for 10 min at 80 °C. The monomer solution was then added to the catalytic solution under dry nitrogen, and the mixture was incubated at 80 °C for 24 h. After the reaction, a small amount of methanol was added to deactivate the catalyst, followed by the addition of toluene until the product was completely dissolved. The product solution was then poured into a large volume of methanol to precipitate the insoluble solids, which were collected by filtration and dried under vacuum. The polymer yield was determined gravimetrically.

2.4. Fabrication of the **Copoly(Flu1-Flu2)** membranes and membrane desilylation

The **Copoly(Flu1-Flu2)** membranes (thickness: 100–220 μm) were fabricated using the casting method. The copolymer solutions (ca. 2.5 wt % in toluene) were filtered to eliminate impurities such as a dust, and then transferred into Petri dishes. The solutions were dried at 25 °C for 96 h to allow slow solvent evaporation, followed by vacuum drying at 60 °C for 12 h to remove residual solvent. After solvent evaporation, the **Copoly(Flu1-Flu2)** membranes were peeled off, immersed in methanol for 24 h, and air-dried at 25 °C to a constant weight.

The **Copoly(Flu1-Flu2)** membranes were immersed in a mixture of hexane and TFA (v/v 1:1) for 48 h, yielded the desilylated membranes, **DScopoly(Flu1-Flu2)**s. The obtained **DScopoly(Flu1-Flu2)** membranes were then washed with triethylamine/hexane, immersed in methanol for 24 h, and air-dried to a constant weight.

2.5. Density of membranes

The measurement of membrane density was performed using a Mettler Toledo balance equipped with a density determination kit, employing the hydrostatic weighing method. The density (ρ) of the membranes was calculated using the equation: $\rho = \frac{\rho_0 M_A}{M_A - M_L}$, where ρ_0 is the density of the aqueous $NaNO_3$ solution (1.16 g/cm^3), M_A is the weight of the membrane in air, and M_L is the weight of the membrane in the aqueous $NaNO_3$ solution.

3. Results and discussion

3.1. Syntheses and characterization of copolymers

The **Copoly(Flu1-Flu2)(4:1)**, **Copoly(Flu1-Flu2)(2:1)**, **Copoly (Flu1-Flu2)(1:1)**, **Copoly(Flu1-Flu-2)(1:2)**, and **Copoly(Flu1-Flu2) (1:4)** were synthesized via copolymerization of **Flu1** with **Flu2** using a $TaCl_5$-n-Bu_4Sn catalytic system at monomer feed ratios of 4:1, 2:1, 1:1, 1:2, and 1:4, respectively. As shown in Table 1, copolymers with varying feed ratios were obtained in moderate yields (64–76 %), indicating efficient polymerization under the specified conditions. Notably, the molecular weights (M_w) of the copolymers exhibited a significant increase with a higher proportion of **Flu2** in the feed, ranging from 629,000 for **Copoly(Flu1-Flu2)(4:1)** to 1,810,000 for **Copoly(Flu1-Flu2)(1:4)**. This trend suggests that **Flu2** likely reduces steric hindrance during the propagation process, as the methyl groups at the 9-position of fluorene in **Flu2** are smaller than the trimethylsilyl groups at the 9-position of fluorene in **Flu1**.

Table 1
Data for the copolymers obtained using the $TaCl_5$-n-Bu_4Sn catalytic system[a].

Feed ratio Flu1:Flu2	Yield (%)[b]	M_w[c]	M_w/M_n[c]
4:1	64	629,000	1.31
2:1	68	744,000	1.27
1:1	74	902,000	1.25
1:2	70	1,050,000	1.49
1:4	76	1,810,000	1.60

[a] Copolymerization conditions: 80 °C, 24 h, $[\mathbf{Flu1} + \mathbf{Flu2}]_0 = 0.20$ M, $[TaCl_5] = 0.020$ M, $[n\text{-}Bu_4Sn] = 0.040$ M.
[b] Methanol-insoluble product.
[c] Measured by GPC.

The ^1H NMR spectra of the **Copoly(Flu1-Flu2)**s, synthesized with varying feed ratios (4:1, 2:1, 1:1, 1:2, and 1:4), are presented in Fig. S1–S5. These spectra consistently exhibit broad peaks at approximately 0.2, 1.4, and 6.5 ppm, which correspond to the trimethylsilyl protons (TMS–H), methyl protons (C–H), and aromatic protons (Ar–H), respectively. The broad nature of these peaks complicates the precise determination of copolymer composition from their integral ratios. Nevertheless, an increasing trend in the integral ratio of TMS–H to C–H is observed as the feed proportion of **Flu1** increases (Fig. S1–S5). The trends observed in the ^1H NMR spectra suggest that the composition ratios of the copolymers approximately correspond to the monomer feed ratios, despite the broad and overlapping signals.

3.2. Membrane fabrication and desilylation

The **Copoly(Flu1-Flu2)** membranes were successfully fabricated through solution casting. Subsequent to desilylation using a hexane/TFA mixture, the **DScopoly(Flu1-Flu2)** membranes exhibited significant

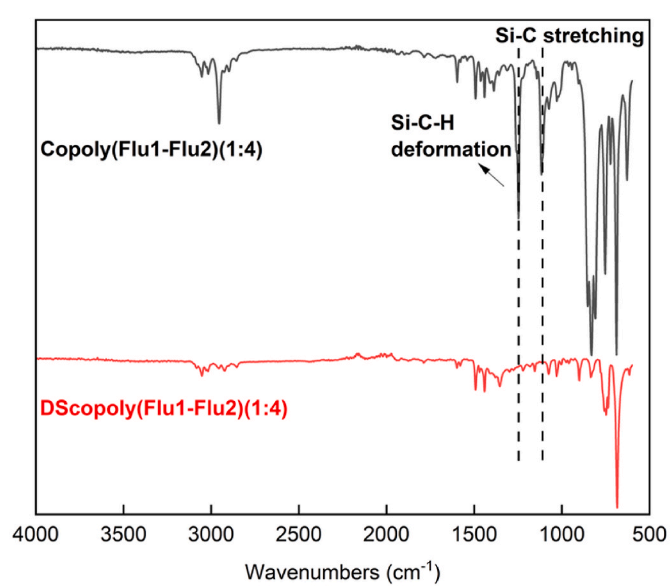


Fig. 1. FT-IR spectra of **Copoly(Flu1-Flu2)(1:4)** and **DScopoly(Flu1-Flu2)(1:4)**.

compositional alterations. The FT-IR spectra of **Copoly(Flu1-Flu2)(1:4)** and **DScopoly(Flu1-Flu2)(1:4)** are presented in Fig. 1. The original copolymer membrane exhibited characteristic absorption bands at 1250 cm^{-1} (Si–C–H deformation) and 1120 cm^{-1} (Si–C stretching), which are attributed to the $SiMe_3$ groups. Following desilylation, these peaks were entirely absent in the **DScopoly(Flu1-Flu2)(1:4)** spectrum, thereby confirming the successful removal of $SiMe_3$ groups. Comparable patterns were observed for other **Copoly(Flu1-Flu2)**s (refer to Fig. S6–S9), indicating that desilylation effectively eradicated silicon-containing functionalities throughout the copolymer series.

The surface morphology and elemental composition of the membranes were further examined using SEM-EDS. Fig. S10 compares the SEM images of **Copoly(Flu1-Flu2)(1:4)** and **DScopoly(Flu1-Flu2)(1:4)**, indicating a slightly rougher surface in the desilylated membrane, likely attributable to microvoid formation following the removal of $SiMe_3$. Fig. S11 and S12 display the EDS spectra and corresponding elemental mapping images of C and Si for the pristine and desilylated membranes, respectively. In **Copoly(Flu1-Flu2)(1:4)** (Fig. S11), distinct Si signals were observed in the mapping images, with silicon uniformly distributed across the membrane surface. Quantitative EDS analysis (Fig. S13) confirmed a significant silicon content of 10.23 wt% in the pristine membrane. In contrast, the desilylated membrane **DScopoly(Flu1-Flu2)(1:4)** (Fig. S12) exhibited a complete absence of Si-related spectral peaks and no detectable silicon in the elemental mapping. This was quantitatively corroborated by EDS data (Fig. S14), which demonstrated a dramatic reduction of silicon content to 0.02 wt %, below the detection limit. The complementary spectral, mapping, and quantitative analyses collectively verify the near-complete elimination of silicon-containing functionalities, consistent with the FT-IR observations.

3.3. *Thermal stability and solubility*

The thermal stability of the membranes was examined using TGA under a nitrogen atmosphere (Fig. 2). All **Copoly(Flu1-Flu2)** and **DScopoly(Flu1-Flu2)** membranes exhibited excellent thermal stability, with minimal weight loss observed below 420 °C. This performance is comparable to other fluorene-containing polymers reported in the literatures [12–14], underscoring the robustness of their backbone structures. Notably, the initial decomposition temperature (T_{onset}) of **Copoly(Flu1-Flu2)** membranes (≈420–430 °C) was slightly lower than that of **DScopoly(Flu1-Flu2)**s (≈450–460 °C), likely due to the earlier degradation tendency of $SiMe_3$ groups in **Copoly(Flu1-Flu2)**s. Further differences in degradation mechanisms appeared above 550 °C. **Copoly(Flu1-Flu2)** membranes showed gradual weight loss at 550–700 °C, with a residual mass of ≈30–50 % at 800 °C. This is likely due to the stepwise decomposition of $SiMe_3$-derived fragments and the fluorene backbone, where silicon-containing residues may retard complete degradation. In contrast, **DScopoly(Flu1-Flu2)** membranes exhibited a steeper weight loss above 550 °C, with residual mass decreasing to ≈ 20–25 % at 800 °C. The absence of $SiMe_3$ groups in desilylated membranes eliminates silicon-based residues, allowing more complete degradation of the fluorene-methyl framework. The exceptional thermal resistance of these materials, attributed to their rigid backbones and strong intermolecular interactions, underscores their suitability for high-temperature gas separation applications.

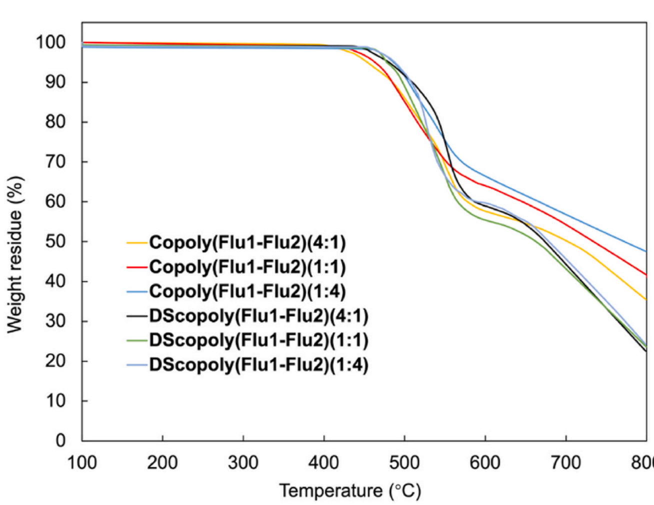


Fig. 2. TGA curves of **Copoly(Flu1-Flu2)** and **DScopoly(Flu1-Flu2)**.

The solubility of the **Copoly(Flu1-Flu2)** and **DScopoly(Flu1-Flu2)** membranes was evaluated in various solvents (Table 2). The pristine **Copoly(Flu1-Flu2)** membranes exhibited solubility in toluene, $CHCl_3$, and THF, but were insoluble in polar solvents such as DMF, DMSO, and CH_3OH, independent of compositions. This phenomenon can be attributed to the presence of a large number of hydrophobic groups in the copolymer molecules, including silyl groups and fluorenyl moieties. These hydrophobic groups enhance the compatibility of the polymers with non-polar or weakly polar solvents. Additionally, the steric hindrance effect of the silyl groups weakens the tight packing between molecular chains, to some extent increasing chain flexibility and making it easier for solvent molecules to penetrate and dissolve the polymers. In contrast, the desilylated **DScopoly(Flu1-Flu2)** membranes became completely insoluble in all tested solvents. This change is likely due to enhanced intermolecular interactions following silicon removal, which restrict chain mobility and reduce solubility. The solubility contrast between as-synthesized and desilylated membranes highlights how desilylation alters molecular forces and chain architecture, thereby influencing dissolution behavior.

3.4. *XRD analysis and N_2 adsorption-desorption results*

The chain packing and microporous structure of the **Copoly(Flu1-Flu2)** and **DScopoly(Flu1-Flu2)** membranes were examined using XRD and N_2 adsorption-desorption measurements. The XRD patterns (Fig. 3 (a) and (b)) exhibited broad, amorphous peaks for all membranes, indicating the absence of long-range crystalline order. **Copoly(Flu1-Flu2)(4:1)** displayed two prominent peaks at 5.8° (d-spacing = 15.2 Å) and 16.3° (d-spacing = 5.4 Å) (Table 3), corresponding to inter-chain packing distances and π-π stacking of aromatic rings, respectively. The notably large d-spacing (15.2 Å) suggests loose chain packing due to the steric hindrance from the bulky $SiMe_3$ groups in **Flu1**. In contrast, **Copoly(Flu1-Flu2)(1:4)** exhibited a peak at 6.4° (d-spacing = 13.8 Å) and 15.2° (d-spacing = 5.8 Å), reflecting slightly denser packing as the proportion of **Flu2** (with smaller 9,9-dimethylfluorenyl groups) increased. After desilylation, **DScopoly(Flu1-Flu2)(4:1)** showed a significant shift of the small-angle peak to 8.4° (d-spacing = 10.5 Å), indicating tighter chain packing due to the removal of the bulky $SiMe_3$ groups. However, **DScopoly(Flu1-Flu2)(1:4)** retains a relatively large d-spacing of 13.4 Å ($2\theta_1 = 6.6°$) because the methyl groups at the 9-position of fluorene in the **Flu2** unit hindered the chain packing. These findings underscore the critical role of chemical structure in regulating chain packing and free volume, which directly influences gas transport properties.

The N_2 adsorption-desorption isotherms (Fig. 3 (c)) confirm the mesoporous nature of the membranes. The pore size distribution curves (Fig. 3 (d)), calculated using the Barrett-Joyner-Halenda (BJH) method,

Table 2
Solubility of **Copoly(Flu1-Flu2)**s and **DScopoly(Flu1-Flu2)**s.

Membrane	Toluene	$CHCl_3$	THF	DMF	DMSO	CH_3OH
Copoly(Flu1-Flu2)s	+	+	+	–	–	–
DScopoly(Flu1-Flu2)s	–	–	–	–	–	–

Symbols: +, soluble; −, insoluble.

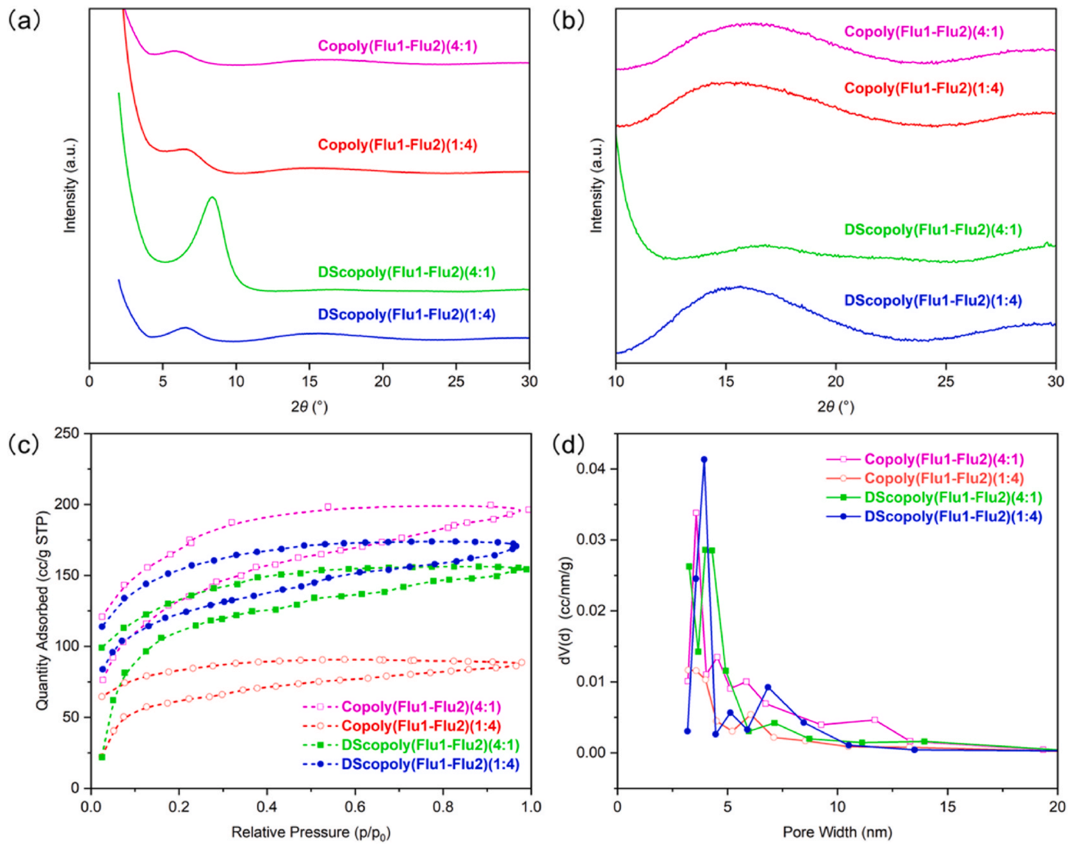


Fig. 3. (a) XRD patterns of membranes; (b) The magnified view of XRD patterns of membranes; (c) N_2 adsorption/desorption isotherm curves at 77 K for membranes; (d) Pore size distribution curves calculated using Barrett-Joyner-Halenda (BJH) for membranes.

Table 3
XRD and N_2 adsorption/desorption data for membranes.

Membrane	$2\theta_1$ [a]	d_1 [b]	$2\theta_2$ [a]	d_2 [b]	S_{BET} [c]	V_{total} [d]
Copoly(Flu1-Flu2)(4:1)	5.8	15.2	16.3	5.4	492	0.299
Copoly(Flu1-Flu2)(1:4)	6.4	13.8	15.2	5.8	298	0.134
DScopoly(Flu1-Flu2)(4:1)	8.4	10.5	16.9	5.2	450	0.238
DScopoly(Flu1-Flu2)(1:4)	6.6	13.4	15.6	5.7	453	0.261

[a] Determined from peak top. In the ∘ unit.
[b] Calculated using equation $\lambda/2\sin\theta$. In the Å unit.
[c] BET surface area calculated from N_2 adsorption isotherm. In the m^2/g unit.
[d] Total pore volume estimated from N_2 uptake at $P/P_0 = 0.96$. In the cm^3/g unit.

reveal multimodal distributions with pore sizes in the range of 3–20 nm for all membranes. The **Copoly(Flu1-Flu2)(4:1)** exhibited the highest BET surface area (492 m^2/g) and total pore volume (0.299 cm^3/g), attributed to its loose chain packing and the presence of large $SiMe_3$ groups, which create abundant voids (Table 3). As the **Flu2** content increased, the BET surface area and total pore volume decreased (e.g., **Copoly(Flu1-Flu2)(1:4)** had 298 m^2/g and 0.134 cm^3/g), consistent with the denser packing observed in XRD. As shown in Fig. 3 (d), a small peak around 12 nm pore width was observed in **Copoly(Flu1-Flu2)(4:1)**, whereas such a peak was not observed in **Copoly(Flu1-Flu2)(1:4)**. Therefore, bulky trimethylsilyl groups at the 9-position of fluorine can play a role in the formation of large pores in the membranes. Desilylation had contrasting effects on the pore structure. **DScopoly(Flu1-Flu2)(4:1)** showed a decrease in BET surface area and total pore volume (450 m^2/g and 0.238 cm^3/g, respectively). From Fig. 3 (d), **DScopoly(Flu1-Flu2)(4:1)** showed no peak attributed to the large pore observed before desilylation. In contrast, **DScopoly(Flu1-Flu2)(1:4)** exhibited an increase in both BET surface area (453 m^2/g) and total pore volume (0.261 cm^3/g), suggesting that the removal of $SiMe_3$ groups generates additional free volume when methyl groups were present at the 9-position of large parts of fluorenes in the copolymers. The size of pores generated by desilylation was estimated to be approximately 7 nm from Fig. 3 (d).

3.5. Gas permeability

The gas permeation characteristics of the synthesized membranes demonstrated significant composition-dependent behavior. The homopolymer **PolyFlu1** [12], derived from 1-phenyl-2-(7-trimethylsilyl-9, 9-bistrimethylsilyl-fluoren-2-yl)acetylene, exhibited high permeability coefficients (P_{O2}: 7000 Barrer), which is consistent with the increased fractional free volume (FFV) imparted by the presence of three bulky $SiMe_3$ groups per repeat unit. Notably, the copolymerization of **Flu1** with **Flu2** significantly enhanced gas permeability. Specifically, **Copoly(Flu1-Flu2)(4:1)** achieved exceptional P_{O2} (19,800 Barrer), surpassing both parent homopolymers (**PolyFlu1** [12] and **PolyFlu2** [13]). This synergistic effect coincided with the lowest membrane density (0.790 g/cm^3), indicating optimized chain packing and maximized FFV at this composition. As the **Flu2** content increased beyond the 4:1 ratio, permeability declined monotonically (Table 4).

Desilylation induced divergent effects on permeability, governed by the initial membrane structure. For homopolymers, **PolyFlu1** exhibited large gas permeability coefficients and low membrane density compared to **DSpolyFlu1**, with desilylation resulting in decreased gas permeability. In contrast, the desilylation of **PolyFlu2** increased gas permeability (e.g., P_{O2}: from 5400 to 9800 Barrer), suggesting efficient microvoid formation, likely due to the presence of methyl groups at the 9-position of fluorenes. The desilylation of **Copoly(Flu1-Flu2)(4:1)**, which consisted of the largest **Flu1** unit, decreased gas permeability. The P_{O2} value of **DScopoly(Flu1-Flu2)(4:1)** was 9000 Barrer,

Table 4
Gas permeability coefficients,[a] and density[b] of membranes.

Membrane	Feed ratio	P_{O2}	P_{N2}	P_{CO2}	P_{O2}/P_{N2}	Density
PolyFlu1[c]	1:0	7000	3700	23000	1.9	0.876
Copoly(Flu1-Flu2)	4:1	19800	15500	36900	1.3	0.790
	2:1	10900	7100	29700	1.5	0.793
	1:1	8700	5500	25400	1.6	0.781
	1:2	8300	5000	23600	1.7	0.801
	1:4	6200	3520	19600	1.8	0.800
PolyFlu2[d]	0:1	5400	3400	18000	1.6	0.810
DSpolyFlu1[c]	1:0	5800	3000	20000	1.9	0.904
DScopoly(Flu1-Flu2)	4:1	9000	5000	26600	1.8	0.757
	2:1	12900	8700	36600	1.5	0.757
	1:1	11800	8000	34100	1.5	0.760
	1:2	10900	6700	32600	1.6	0.769
	1:4	10300	6500	31800	1.6	0.783
DSpolyFlu2[d]	0:1	9800	8100	24000	1.2	0.770

[a] Determined at 25 °C. Units: Barrer, 1 Barrer = 10^{-10} cm^3 (STP) cm cm^{-2} s^{-1} $cmHg^{-1}$.
[b] Determined by hydrostatic weighing. Units: g/cm^3.
[c] Data from Ref. [12].
[d] Data from Ref. [13].

significantly lower than the 19,800 Barrer of **Copoly(Flu1-Flu2)(4:1)**. This result can be anticipated from the variation in BET surface area and total pore volume described above. The other **DScopoly(Flu1-Flu2)** membranes displayed enhanced permeability, with the 2:1 ratio showing particularly notable improvements (P_{O2} from 10,900 to 12,900 Barrer).

All membranes exhibited moderate selectivity (P_{O2}/P_{N2}: 1.2–1.9), characteristic of high-free-volume glassy polymers where permeability-selectivity trade-offs prevail. A strong inverse correlation was observed between gas permeability and membrane density, with **Copoly(Flu1-Flu2)(4:1)** achieving the highest permeability concurrent with the lowest density (0.790 g/cm^3). Desilylation further reduced densities for most **DScopoly(Flu1-Flu2)** membranes (e.g., 0.757 g/cm^3 for 4:1 feed ratios), aligning with their elevated permeabilities.

3.6. Physical aging

Physical aging is a critical issue for glassy polymers, as it results in a gradual reduction in gas permeability over time due to the relaxation of polymer chains towards a more thermodynamically stable state. The aging behavior of the membranes was evaluated by monitoring oxygen permeability over a 240-day period, with the results depicted in Fig. 4 and the specific change values are shown in Table S1. Among the homopolymers, **PolyFlu1** exhibited a notable decrease in P_{O2} from 7000 Barrer to 5300 Barrer over 240 days, representing a 24.3 % reduction. This decline is attributed to the relaxation of polymer chains. Conversely, **PolyFlu2** showed a more moderate aging effect, with a 9.3 % reduction in P_{O2} (5400 to 4900 Barrer). Desilylation of **PolyFlu1** (**DSpolyFlu1**) revealed significant physical aging, with P_{O2} decreasing by 25.9 % (5800 to 4300 Barrer), while **DSpolyFlu2** showed a 9.2 % reduction (9800 to 8900 Barrer), suggesting that **Flu2**-based structures mitigated chain relaxation.

The **Copoly(Flu1-Flu2)** and **DScopoly(Flu1-Flu2)** membranes exhibited exceptional aging resistance, with permeability losses below 10 % over 240 days for optimized compositions. This behavior is attributed to their rigid backbones, tailored free volume, and the synergistic effects of copolymerization and desilylation. Mechanistically, aging is governed by free volume dynamics and chain mobility. Polymers with higher free volume, such as **Flu1**-rich copolymers, exhibit higher initial permeabilities but are more susceptible to aging due to large excess free volume and good chain mobility of $SiMe_3$. Desilylation affects aging differentially: in **Flu1**-rich systems, it reduces free volume and promotes chain packing due to the absence of bulky substituents at the 9-position of fluorenes, accelerating aging, whereas in **Flu2**-rich systems, it creates a rigid framework that hinders relaxation due to the presence of bulky methyl groups at the 9-position of fluorenes. The **Flu1**:**Flu2** ratio of 2:1 in both copolymers and desilylated membranes strikes a balance between permeability and aging resistance, rendering it a promising candidate for gas separation applications. These findings underscore the importance of molecular design in tailoring polymer structures to minimize physical aging while maintaining high gas transport performance.

4. Conclusions

In this study, silyl-functionalized fluorene-based poly(substituted acetylene)s (**Copoly(Flu1-Flu2)**s) were synthesized successfully through copolymerization using a $TaCl_5$-n-Bu_4Sn catalytic system, with the feed ratios of **Flu1** and **Flu2** systematically adjusted to optimize polymer structures. The copolymers demonstrated high molecular weights, forming free-standing membranes via solution casting. Desilylation with trifluoroacetic acid effectively removed $SiMe_3$ groups, as confirmed by FT-IR and SEM-EDS analyses. The **Copoly(Flu1-Flu2)** membranes and the desilylated **DScopoly(Flu1-Flu2)** membranes exhibited excellent gas transport properties (e.g., **Copoly(Flu1-Flu2)(4:1)** showed P_{O2} of 19,800 Barrer), along with good thermal stability (decomposition temperature >420 °C) and resistance to physical aging (P_{O2} reduction <10 % within 240 days). XRD analysis indicated loose interchain packing (d-spacing 13.4–15.2 Å), and N_2 adsorption revealed high BET surface areas (up to 492 m^2/g) and total pore volumes (0.299 cm^3/g). These findings underscore the critical role of steric hindrance from fluorenyl substituents and desilylation-induced porosity in

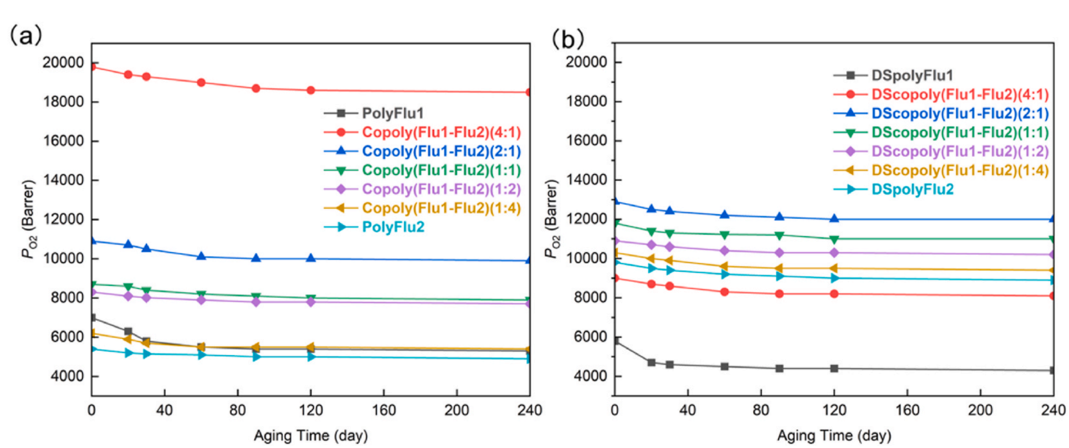


Fig. 4. Effect of physical aging on the P_{O2} of (a) membranes before desilylation; (b) membranes after desilylation.

achieving superior gas transport, offering new strategies for high-performance membrane design in industrial gas separation.

CRediT authorship contribution statement

Yi Lin: Writing – original draft, Validation, Resources, Investigation, Funding acquisition, Data curation, Conceptualization. **Toshikazu Sakaguchi:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Tamotsu Hashimoto:** Writing – review & editing, Supervision, Resources. **Xiaoli Sun:** Validation, Investigation, Data curation. **Yanbing Guo:** Validation, Investigation, Formal analysis. **Jiping Chen:** Validation, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124830.

Data availability

No data was used for the research described in the article.

References

[1] Y. Zhang, B.H. Yin, L. Huang, et al., MOF membranes for gas separations, Prog. Mater. Sci. 151 (2025) 101432.
[2] S. Li, C. Ma, J. Hou, et al., Highly porous metal-organic framework glass design and application for gas separation membranes, Nat. Commun. 16 (2025) 1622.
[3] B. Qiu, Y. Gao, P. Gorgojo, et al., Membranes of polymer of intrinsic microporosity PIM-1 for gas separation: modification strategies and meta-analysis, Nano-Micro Lett. 17 (2025) 114.
[4] A. Mollahosseini, M.N. Dafchahi, S.K. Salestan, et al., Polymeric membranes in carbon capture, utilization, and storage: current trends and future directions in decarbonization of industrial flue gas and climate change mitigation, Energy Environ. Sci. 18 (2025) 5025–5092.
[5] D. Borah, G. Hazarika, A. Gogoi, et al., Polymeric membranes for sustainable gas separation: a comprehensive review with challenges, innovations and future perspectives, Renew. Sustain. Energy Rev. 219 (2025) 115868.
[6] B. Comesaña-Gándara, J. Chen, C.G. Bezzu, et al., Redefining the Robeson upper bounds for CO_2/CH_4 and CO_2/N_2 separations using a series of ultrapermeable benzotriptycene-based polymers of intrinsic microporosity, Energy Environ. Sci. 12 (2019) 2733–2740.
[7] L. Sun, W. Xu, H. Zhang, et al., In-situ formation of three-dimensional network intrinsic microporous ladder polymer membranes with ultra-high gas separation performance and anti-trade-off effect, Angew. Chem. Int. Ed. 64 (2025) e202420742.
[8] M. Chen, G. Hu, T. Shen, et al., Applications of polyacetylene derivatives in gas and liquid separation, Molecules 28 (2023) 2748.
[9] Y. Lin, T. Sakaguchi, T. Hashimoto, Ultrahigh oxygen permeability of the desilylated membranes of halogen-containing diphenylacetylene copolymers, J. Membr. Sci. 628 (2021) 119272.
[10] Y. Lin, T. Sakaguchi, T. Hashimoto, Imidazolium-based diphenylacetylene copolymers with excellent carbon dioxide/nitrogen and oxygen/nitrogen gas separation performance, J. Membr. Sci. 637 (2021) 119638.
[11] T. Sakaguchi, M. Kuroiwa, T. Hashimoto, Gas-permeable fluorine-substituted poly (diphenylacetylene)s: effect of number and position of fluorine atoms on gas permeability, Polymer 283 (2023) 26264.
[12] Y. Lin, T. Sakaguchi, T. Hashimoto, Synthesis of poly[1-phenyl-2-(fluoren-2-yl) acetylene]s bearing trimethylsilyl groups at the fluorene 9-position to give highly gas-permeable polymer membranes, Polymer 230 (2021) 124087.
[13] A. Fukui, K. Hattori, Y. Hu, et al., Synthesis, characterization, and high gas permeability of poly(diarylacetylene)s having fluorenyl groups, Polymer 50 (2009) 4159–4165.
[14] Y. Hu, K. Hattori, A. Fukui, et al., Synthesis and gas permeation properties of various Si-containing poly(diarylacetylene)s and their desilylated membranes, Polymer 51 (2010) 1548–1554.
[15] Y. Lin, T. Sakaguchi, T. Hashimoto, Extremely high gas permeability of naphthyl group-containing diarylacetylene copolymers, Polymer 212 (2021) 123305.
[16] Y. Lin, D. Rong, T. Sakaguchi, et al., Development of highly gas-permeable polymers by metathesis copolymerization of 1-(p-trimethylsilyl) phenyl-1-propyne with tert-butyl and silyl group-containing diphenylacetylenes, RSC Adv. 11 (2021) 32419–32424.
[17] T. Sakaguchi, K. Kuratani, T. Hashimoto, Effect of methyl group on gas permeability of trimethylsilyl-containing poly (diphenylacetylene)s, Polymer 255 (2022) 125138.
[18] Z.X. Low, P.M. Budd, N.B. McKeown, et al., Gas permeation properties, physical aging, and its mitigation in high free volume glassy polymers, Chem. Rev. 118 (2018) 5871–5911.
[19] F. Almansour, A.B. Foster, A.W. Ameen, et al., High gas permeance in CO_2-selective thin film composite membranes from bis(phenyl)fluorene-containing blends with PIM-1, J. Membr. Sci. 699 (2024) 122652.
[20] Z. Li, Y. Meng, N. Zhang, et al., Crosslinked microporous membrane with pore compensation for efficient and long-term gas separation, J. Membr. Sci. 711 (2024) 123201.
[21] S.J.D. Smith, R. Hou, K. Konstas, et al., Control of physical aging in super-glassy polymer mixed matrix membranes, Acc. Chem. Res. 53 (2020) 1381–1388.
[22] D.S. Bakhtin, S.E. Sokolov, I.L. Borisov, et al., Mitigation of physical aging of polymeric membrane materials for gas separation: a review, Membranes 13 (2023) 519.
[23] M.Z. Ahmad, R. Castro-Muñoz, P.M. Budd, Boosting gas separation performance and suppressing the physical aging of polymers of intrinsic microporosity (PIM-1) by nanomaterial blending, Nanoscale 12 (2020) 23333–123370.
[24] T.J. Corrado, Z. Huang, D. Huang, et al., Pentiptycene-based ladder polymers with configurational free volume for enhanced gas separation performance and physical aging resistance, Proc. Natl. Acad. Sci. USA 118 (2021) e2022204118.